Low Complexity Global Motion Estimation from Block Motion Vectors

Stanislav Soldatov, Konstantin Strelnikov, Dmitriy Vatolin
Moscow State University, department of Computational Mathematics and Cybernetics,
Graphics and Media Laboratory
{stass, kstrelnikov,dmitriy}@graphics.cs.msu.ru

Abstract

For practical applications there is a requirement of the global motion estimation that could use existing block matching motion data. In this paper authors propose two global motion estimation methods that are based on block matching motion estimation. These methods estimate different global motion models. One of the proposed method advantages consists in motion field filtration that allows coping with low reliability of motion vectors. In particular method of motion vectors reliability estimation is proposed. Model parameters are estimated sequentially, allowing to perform additional filtration to increase robustness. Proposed algorithms show good results in terms of robustness and precision while maintaining high processing speed and reasonable memory requirements.

Keywords: *global motion estimation, motion vector reliability estimation, motion field filtration.*

1 INTRODUCTION

Global motion estimation task is employed in many current applications and is investigated for a long time. Existing global motion estimation algorithms are used for video compression [Keller and Averbuch 2003], video segmentation, video indexing, panoramic image construction (mosaicing) and video stabilization.

Global motion estimates translation from one frame to another. Motion model defines type of translation that is estimated when global motion estimation problem is solved. Model complexity depends on field of application. In most cases two-, four-, six- and eight-parameter translation models are used.

When motion model is defined then global motion estimation problem is reduced to model parameter estimation problem. Three main approaches to model parameters estimation exist with specific advantages and drawbacks.

1) Methods based on feature points.
2) Methods based on block motion vectors.
3) Global search methods.

Methods based on feature points have high computational complexity, because feature selection and feature tracking are computationally expensive operations [Shi and Tomasi 1999]. Advantage of algorithms from first group is high reliability of obtained shifts. Reliability of shifts is obtained using block matching motion estimation algorithms is worse [Dufaux and Moscheni 1995]. But this approach has strong advantages including easiness and high computing speed. Global search method group contain algorithms which process whole frame without auxiliary transformation [Litvin et al. 2003]. Computational complexity of these algorithms is higher than computational complexity of algorithms based on feature points.

Dwell on commonly used motion models. Two-parameter motion model is most simple. It describes only horizontal and vertical shifts. Two-parameter motion model is usually used in fast video stabilization. Two-parameter model estimation with algorithms based on bit-plane matching is described in [Lee et al. 1998]. Four-parameter motion model describes horizontal and vertical shifts, rotation and zoom. This model can be used in mosaicing and video stabilization tasks [Morimoto and Chellappa 1996]. Sometimes rotation parameter is not described [Chiew et al. 2004]. Six-parameter motion model is defined as affine translation. It is usually used in video coding [Chiew et al. 2004] and video stabilization tasks [Litvin et al. 2003]. Eight-parameter model is most precise from all models described here. This model describes perspective transformation and usually is used in accurate video mosaicing [Szelski 1996].

In this paper, we propose two high quality global motion estimation algorithms based on conventional block motion vectors. First algorithm estimates four-parameter motion model and other estimates six-parameter motion model. Algorithms use high-speed approach of motion estimation and minimize errors of global motion estimation introduced by low reliable motion vectors. Here authors introduce the use of several block motion vector reliability estimation techniques.

2 ERRONEOUS MOTION VECTOR FILTRATION

2.1 Defects of block matching motion estimation

Motion estimation is important problem in video coding and video processing. There are different approaches to motion estimation namely pixel-based, block-based, feature-based, model-based and object-based algorithms. Block-based algorithms (Block Matching) most commonly used due to their simplicity. Exist quick algorithms of block matching motion estimation. Therefore here authors chosen block matching as basis of proposed global motion estimation algorithms.

Full search block matching is best from block matching algorithms. But this method has highest computational complexity, so it is seldom used in real applications. The idea of proposed algorithms is low complexity in combination with high precision. As a base authors chose fast motion estimation algorithm called - 3D Recursive Search (3DRS) [Olivieri et al. 1999] with some changes. Even full search algorithm does not always find reliable motion vectors. All the more that is right for fast method.

Motion field, estimated by block matching, has some typical features.

1) If SAD of motion vector is high, most probably the motion vector is erroneous, because associated blocks have low resemblance.

2) If motion vector strongly differs from neighboring vectors, most probably this motion vector is erroneous. That is consequence of motion field smoothness.
3) If dispersion of motion vector's block is low, most probably the motion vector is erroneous.

Figure 1 shows the example of erroneous motion vectors.

Motion field   Residual after compensation

Figure 1: Erroneous motion vectors (frame 9 of "foreman" movie). Blue vectors correspond to low dispersion blocks. Red vectors have high SAD.

2.2 Motion field filtration based on motion vectors' reliability estimation

Proposed global motion estimation methods do not require whole motion field but accuracy of vectors is important for them. Therefore erroneous motion vectors filtration becomes very important for correct estimation.

Authors propose function that estimates degree of conformity a motion vector and real motion – belief function. This function employs expert knowledge on typical block motion estimation problems.

Belief function depends on following characteristics:

1) $error_{x,y}$ – approximation error (sum of absolute difference of pixels from current frame's block in (x,y) position and corresponding pixels from previous frame);
2) $disp_{x,y}$ – dispersion of current frame's block in (x,y) position;
3) $dev_{x,y}$ – mean square deviation of vectors from neighbors (see formula 1).

$$dev_{x,y} = \frac{1}{4} \cdot \sum_{\substack{i=-1;1 \\ j=-1;1}} \begin{bmatrix} \left(mv^X_{x,y} - mv^X_{x+i,y+j}\right)^2 + \\ \left(mv^X_{x,y} - mv^Y_{x+i,y+j}\right)^2 \end{bmatrix} \tag{1}$$

$mv^X_{x,y}$, $mv^Y_{x,y}$ – motion vector projections onto abscissa and ordinate axis.

Formula (2) shows proposed belief function. We use it to motion field filtration next manner. Vectors with low belief values are discarded and all other vectors are kept.

$$belief_{x,y} = \left(a \cdot error_{x,y} + \frac{b}{disp_{x,y}^{\ 2}} + c \cdot dev_{x,y}\right)^{-1} \tag{2}$$

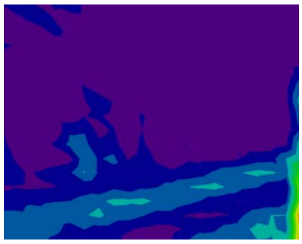
a,b,c – estimated parameters.

Authors conducted experiments on sequence database that gave following belief function parameter values: a=0.25; b=32; c=1.

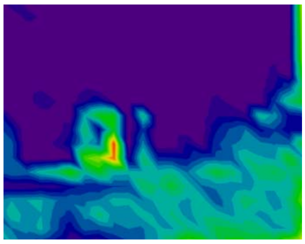
Belief function on figure 2 is typical example of motion vectors reliability estimation. Figure 2 shows characteristics of motion field and estimated belief values. Belief values descend from dark blue areas to green, further to yellow and red. On figure 2 it could be seen that vectors corresponding to low values of belief function are incorrect (red areas on figure 2).

Further we use only vectors with belief function values higher than a threshold.

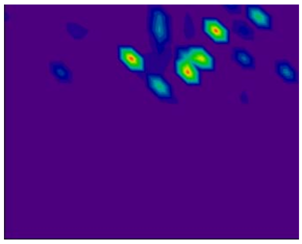
Frame with motion vectors



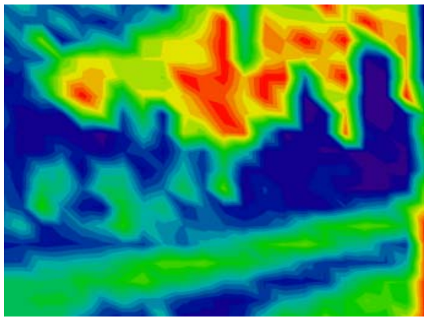
Approximation error of blocks



Block dispersion



Mean square deviation



Belief function

Figure 2: Motion vectors' reliability estimation (frame 3 of "flower" movie).

3 GLOBAL MOTION ESTIMATION

Information about shifts of frame's slices allows estimating global motion of frame. Different motion models can be used to approximate global motion. To describe transformation between frames I' and I'' following general model could be used:

$$\begin{pmatrix} x'' \\ y'' \end{pmatrix} \sim \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix} + \begin{pmatrix} A_{13} \\ A_{23} \end{pmatrix} \tag{3}$$

Where

A_{ij} ($i \in [1;2]$, $j \in [1;3]$) – model parameters (some parameters may be defined as constant values);

$(x';y')$ – pixels of frame I';

$(x'';y'')$ – corresponding pixels of frame I''.

Global motion estimation task is formulated as minimization problem of function of six parameters (4).

$$F(I',I'';A_{11},..,A_{23}) = \sum_{i,j}\begin{bmatrix}\left(A_{11}\cdot x'_{i,j}+A_{12}\cdot y'_{i,j}+A_{13}-x''_{i,j}\right)^2 + \\ \left(A_{21}\cdot x'_{i,j}+A_{22}\cdot y'_{i,j}+A_{23}-y''_{i,j}\right)^2\end{bmatrix} \quad (4)$$

In this paper algorithms of global motion estimation for two specific models are described:

1) four-parameter model

$$I''=\begin{pmatrix} s\cdot\cos\varphi & -s\cdot\sin\varphi \\ s\cdot\sin\varphi & s\cdot\cos\varphi\end{pmatrix}I'+\begin{pmatrix}a\\b\end{pmatrix}=F(a,b,s,\varphi)I' \quad (5)$$

a,b,s,φ – model parameters: horizontal shift, vertical shift, zoom and rotate.

2) six-parameter model

$$I''=\begin{pmatrix} a & b \\ d & e\end{pmatrix}I'+\begin{pmatrix}c\\f\end{pmatrix}=F(a,b,c,d,e,f)I' \quad (6)$$

a,b,c,d,e,f – model parameters.

Decision on which model should be used depends from motion estimation application. Four-parameter model is more robust, but imposes constraints on input data, that could be not met for some applications.

3.1 Four-parameter motion model estimation method

3.1.1 Estimation of Zoom Parameter

High-precision estimation of parameter s value is very important for four-parameter model. Wrong estimation can introduce a strong error into other model parameters. Authors suggest following calculation algorithm of s parameter.

Parameter s estimation problem is solved easily and accurately for any given motion vector pair.

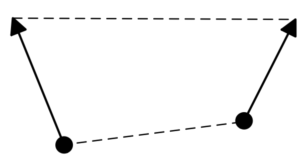


Figure 3: s parameter estimation for motion vector pair.

Let consider pair of vectors from figure 3. For them formula (7) is correct.

$$\begin{pmatrix}x_i'\\y_i'\end{pmatrix}=\begin{pmatrix}s_{V1,V2}\cdot\cos\varphi & -s_{V1,V2}\cdot\sin\varphi\\ s_{V1,V2}\cdot\sin\varphi & s_{V1,V2}\cdot\cos\varphi\end{pmatrix}\cdot\begin{pmatrix}x_i\\y_i\end{pmatrix}+\begin{pmatrix}a\\b\end{pmatrix} \quad i=1,2 \quad (7)$$

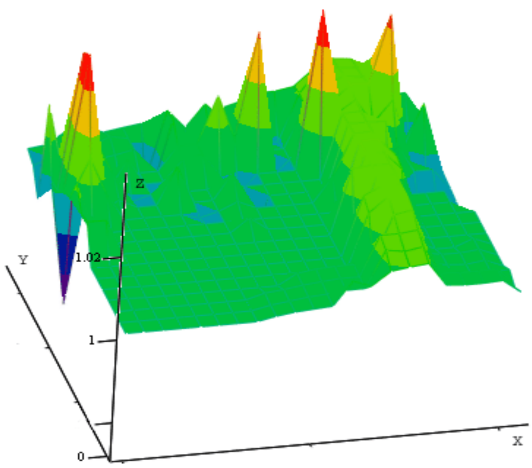
After several transformations we get formula (8) for parameter s calculation for two known vectors.

$$s_{V1,V2}=\sqrt{\frac{(x_2'-x_1')^2+(y_2'-y_1')^2}{(x_2-x_1)^2+(y_2-y_1)^2}} \quad (8)$$

We use this fact in our algorithm. Get N random vectors from set of reliable vectors. For each selected vector V_n ($n \in 1..N$) parameter z_n is calculated that is local zoom value for this motion vector. For that random vectors mv_m^n ($m \in 1..M$) are selected from set of reliable vectors. Values $s_{n,m}$ are calculated for each pair of vectors V_n and mv_m^n ($n \in 1..N$, $m \in 1..M$) using formula (8). Value z_n is calculated as argument of histogram maximum (see formula 9).

$$z_n = \arg\left(\max\left(\underset{m\in[1;M]}{hist}(s_{n,m})\right)\right),\ n \in 1..N \quad (9)$$

Frame 210 of “flower” movie with motion vectors



z_n values of motion vectors

Figure 4: Additional filtration of motion vectors.

Parameter s is estimated from a formula, similar to (9) as argument of histogram of z_n ($n \in 1..N$) values maximum.

Some vectors of motion field can correspond with objects at a short distance or moving objects (e.g. Figure 4). These vectors should not be used in global motion estimation algorithm because they introduce an error.

Figure 4 shows that problem vectors can be filtered using z_n values obtained for these vectors. Filtration is performed in following manner. From set of vectors V_n ($n \in 1..N$) are kept vectors VF_k ($k \in 1..K, K < N$) whose parameter z_k belongs to neighborhood of s, i.e. $z_k \in \left[s-t; s+t\right]$. t is experimentally defined threshold, that was calculated under the requirement that at least 50% of reliable vectors must be kept.

3.1.2 Rotation and shifts estimation

We use least-squares method to estimation of rotation and shift parameters of model. Formula (10) defines loss function $F(a,b,\varphi)$.

$$F(a,b,\varphi) = \sum_{k\in[1;K]} \left[\begin{array}{l} \left(s\cdot\cos\varphi\cdot VF^{X1}{}_k - s\cdot\sin\varphi\cdot VF^{Y1}{}_k + a - VF^{X2}{}_k\right)^2 + \\ \left(s\cdot\sin\varphi\cdot VF^{X1}{}_k + s\cdot\cos\varphi\cdot VF^{Y1}{}_k + b - VF^{Y2}{}_k\right)^2 \end{array} \right] \quad (10)$$

- VF^{X1}, VF^{Y1} –vector origin coordinates ;
- VF^{X2}, VF^{Y2} –vector terminus coordinates.

For parameters calculation loss function is differentiated by each parameter. Result is equated with zero and obtained equations set is solved.

As a result we get formulas (11-13) to calculation a, b and φ parameters of model.

$$\varphi = \arccos\left(\begin{array}{l} K\cdot \sum_{k\in[1;K]} VF^{X2}{}_k{}^2 + K\cdot \sum_{k\in[1;K]} VF^{Y2}{}_k{}^2 - \\ \left[\sum_{k\in[1;K]} VF^{X2}{}_k\right]^2 - \left[\sum_{k\in[1;K]} VF^{Y2}{}_k\right]^2 \end{array} \right) + \\ \arccos\left(s\cdot \left[\begin{array}{l} \sum_{k\in[1;K]} VF^{X1}{}_k \cdot \sum_{k\in[1;K]} VF^{X2}{}_k + \sum_{k\in[1;K]} VF^{X1}{}_k \cdot \sum_{k\in[1;K]} VF^{Y2}{}_k - \\ \sum_{k\in[1;K]} VF^{Y1}{}_k \cdot \sum_{k\in[1;K]} VF^{X2}{}_k + \sum_{k\in[1;K]} VF^{Y1}{}_k \cdot \sum_{k\in[1;K]} VF^{Y2}{}_k - \\ -K\cdot \sum_{k\in[1;K]} VF^{X1}{}_k \cdot VF^{X2}{}_k - K\cdot \sum_{k\in[1;K]} VF^{X1}{}_k \cdot VF^{Y2}{}_k + \\ K\cdot \sum_{k\in[1;K]} VF^{X1}{}_k \cdot VF^{Y2}{}_k - K\cdot \sum_{k\in[1;K]} VF^{Y1}{}_k \cdot VF^{Y2}{}_k \end{array} \right] \right) \quad (11)$$

$$a = \frac{1}{K}\cdot\left(\begin{array}{l} \sum_{k\in[1;K]} VF^{X2}{}_k - s\cdot\cos\varphi\cdot \sum_{k\in[1;K]} VF^{X1}{}_k + \\ s\cdot\sin\varphi\cdot \sum_{k\in[1;K]} VF^{Y1}{}_k \end{array} \right) \quad (12)$$

$$b = \frac{1}{K}\cdot\left(\begin{array}{l} \sum_{k\in[1;K]} VF^{Y2}{}_k - s\cdot\sin\varphi\cdot \sum_{k\in[1;K]} VF^{Y1}{}_k - \\ s\cdot\cos\varphi\cdot \sum_{k\in[1;K]} VF^{Y1}{}_k \end{array} \right) \quad (13)$$

Figure 5 shows block diagram of proposed four-parameter global motion estimation algorithm.

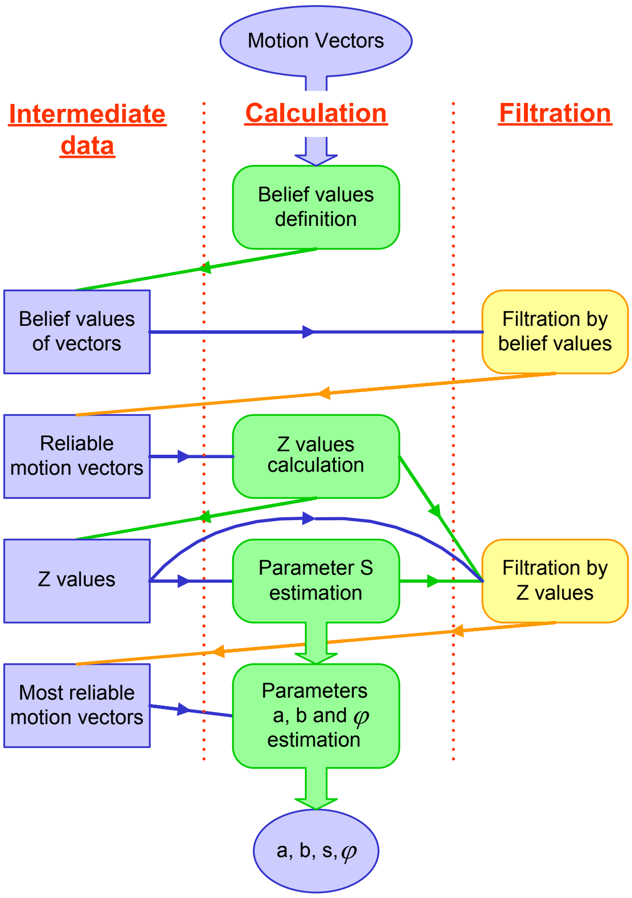


Figure 5: Block-diagram of four-parameter global motion estimation algorithm.

Separated calculation of S and other three model values allows using S value in additional data filtration. Such technique gives this algorithm significant advantage in robustness.

3.2 Six-parameter motion model estimation method

3.2.1 Local parameters estimation

For given three motion vectors parameter values could be calculated unambiguously.

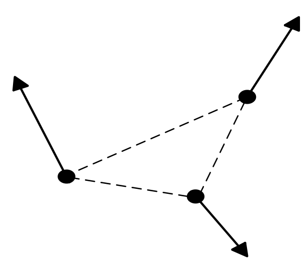


Figure 6: Calculation of parameters for three vectors.

Let’s examine three vectors from figure 6. All three of them are subject of transformation (6). So the following formula (14) is correct.

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \times \begin{pmatrix} a & d \\ b & e \\ c & f \end{pmatrix} = \begin{pmatrix} x_1' & y_1' \\ x_2' & y_2' \\ x_3' & y_3' \end{pmatrix} \tag{14}$$

From formula (14) is easily obtained formula (15) for parameter values calculation.

$$\begin{pmatrix} a & d \\ b & e \\ c & f \end{pmatrix} = \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}^{-1} \times \begin{pmatrix} x_1' & y_1' \\ x_2' & y_2' \\ x_3' & y_3' \end{pmatrix} \tag{15}$$

To estimate selected vector parameters two other random vectors are chosen from reliable set. Local parameters are computed using formula (15).

3.2.2 Estimation of six-parameters model’s parameters

Let's choose N vectors from reliable vector set. For each reliable vector M random pairs of vectors are chosen from a reliable set and M local parameters are calculated for them. Thus we get set of $M \times N$ parameter vectors, where each vector consists of six local parameters.

Global motion parameters are estimated from this set in five steps:

Step 1. Two-dimensional histogram is built for a and c parameters from parameter vectors. Histogram maximum value is found. Values of parameters a and c are defined, as arguments of found maximum.

Step 2. Next step is filtering of parameter vectors in set by throwing away vectors with a and c components too distinct from obtained a and c global motion parameter values.

Step 3. Two-dimensional histogram is built for e and f parameters from current parameter vector set. Values of global parameters are defined similarly as in step 1.

Step 4. Filtering parameter vector set by throwing away vectors with e and f components too distinct from obtained e and f global motion parameter values.

Step 5. Parameters b and d are calculated as in Step 1.

This method of pair-wise estimation and filtration allows improving precision of obtained model parameters significantly. Sequential histogram filtration also allows this algorithm to stay in reasonable memory requirements range, which would be not possible if all parameters were found using one six-dimensional histogram.

4 RESULTS

We compare proposed algorithm of four-parameter global motion estimation and least-squares method without filtration of motion vectors using belief function. Least-squares method gives formulas (16-20) to calculation of parameters a, b, s, φ :

$$a = (X' - c \cdot X + d \cdot Y) / N \tag{16}$$

$$b = (Y' - c \cdot Y - d \cdot Y) / N \tag{17}$$

$$s = \sqrt{c^2 + d^2} \tag{18}$$

$$\varphi = arctg\,(d / c) \tag{20}$$

Where

$$c = \frac{N \cdot XX' - X \cdot X' + N \cdot YY' - Y \cdot Y'}{N \cdot XX - X^2 - N \cdot YY + Y^2} \tag{21}$$

$$d = \frac{N \cdot XY' - X \cdot Y' - N \cdot YX' + Y \cdot X'}{N \cdot XX - X^2 + N \cdot YY - Y^2} \tag{22}$$

$$N = \sum_{x,y} 1;\; X = \sum_{x,y} x;\; Y = \sum_{x,y} y;$$

$$X' = \sum_{x,y} (x + mv_{x,y}^X);\; Y' = \sum_{x,y} (y + mv_{x,y}^Y);$$

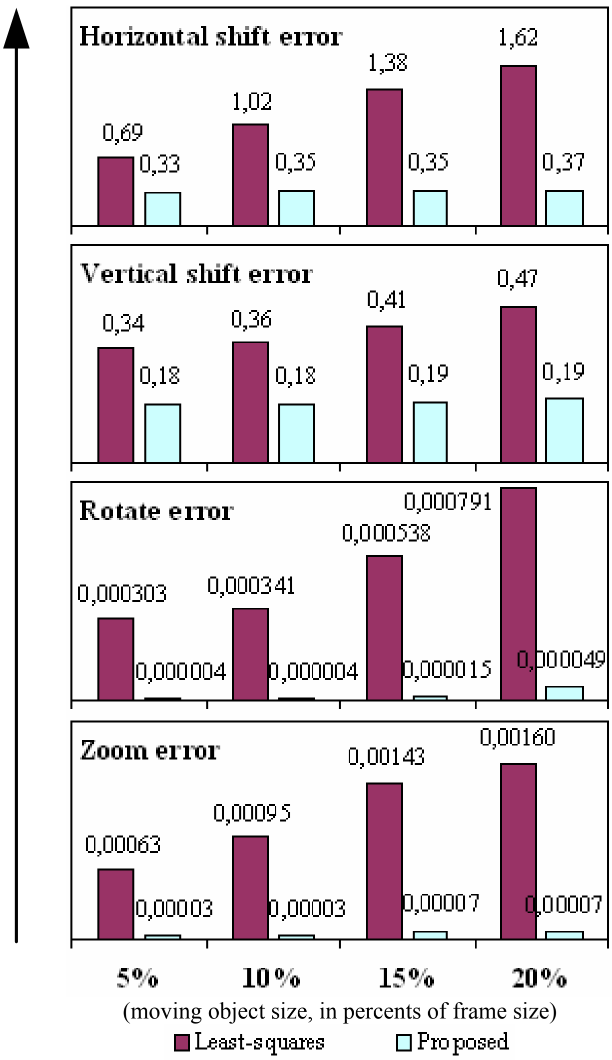


Figure 7: Least-squares and proposed four-parameter method global motion parameters estimation error.

$$XX' = \sum_{x,y} x \cdot (x + mv^X_{x,y})\ ;\ YY' = \sum_{x,y} y \cdot (y + mv^Y_{x,y})\ ;$$

$$XY' = \sum_{x,y} x \cdot (y + mv^Y_{x,y})\ ;\ YX' = \sum_{x,y} y \cdot (x + mv^X_{x,y})\ ;$$

$mv^X_{x,y}; mv^Y_{x,y}$ - projection of motion vector in (x, y) position onto abscissa axis and ordinate axis correspondingly.

4.1 Numerical comparison of estimation accuracy

Numerical comparison is implemented using synthetic video sequences. Real global motion is known for them. In addition an object with different moving direction was added in synthetic video. Measured results are presented on figure 7. They show average error of least-squares method and proposed four-parameter method global motion parameters estimation depending on added moving object size.

Statistics show that proposed algorithm greatly excels the least-squares method in global motion estimation robustness and accuracy.

4.2 Visual comparison

Panorama image construction is used to make visual comparison of algorithms. One algorithm of panorama construction is used. Authors conducted experiments on vast sequence database. Figure 8 is typical result of algorithms work. It allows to compare quality of global motion estimation. The figure shows two panoramas are constructed using global motion parameters from single sequence. Top picture is a result of panorama construction using parameters by least-squares method (16-20). Bottom picture is a result of panorama construction using parameters that are obtained by proposed algorithm (9,11-13). It is can be seen that proposed algorithm gives better result than the least-squares method.

Least-squares method

Proposed four-parameter algorithm

Figure 8: Comparison of algorithms ("bus" movie).

4.3 Processing speed

Table 1 and table 2 show average frame processing time and FPS of proposed algorithms on P-IV 1.6GHz.

Table 1. Four-parameter model estimation frame processing time for CIF resolution movie on Pentium-IV 1.6GHz.

	Time without ME		Time with ME	
Proposed	4.5 ms	219 fps	36.8 ms	27 fps
Least-squares	1.3 ms	770 fps	33.6 ms	30 fps

Table 2. Six-parameter model estimation frame processing time for CIF resolution movie on Pentium-IV 1.6GHz.

	Time without ME		Time with ME	
Proposed	10.8 ms	92.5 fps	43.1 ms	23 fps
Least-squares	1.3 ms	770 fps	33.6 ms	30 fps

5 CONCLUSION

Authors propose methods that robustly estimate two global motion models with high precision. Authors propose motion vector filtration scheme that allows to increase robustness while using existing, possibly hardware implemented, motion estimation algorithms. Proposed algorithms have real-time processing speed even when run simultaneously with software-implemented motion estimation algorithm. Both methods also require reasonable amounts of memory for processing.

Measurements show that proposed methods greatly outperform least-square estimation in precision without losing much in terms of speed.

This makes proposed algorithms suitable for many actual applications, for example panorama construction and video stabilization tasks.

6 REFERENCES

CHIEW, T.-K., HILL, P., BULL, D.R., CANAGARAJAH C.N. 2004. "Robust global motion estimation using the Hough transform for real time video coding". *Picture Coding Symposium 2004*.

DUFAUX, F., KONRAD, J. 2000. "Efficient, robust, and fast global motion estimation for video coding". In *IEEE Transactions on Image Processing*, *Vol. 9*, pp. 497-501.

DUFAUX, F., MOSCHENI, F. 1995. "Motion estimation techniques for digital TV: A review and a new contribution". In *Proceedings of IEEE*, *Vol. 83*, pp. 858-876.

KELLER, Y., AVERBUCH, A. 2003. "Fast Global Motion Estimation for MPEG-4 Video Compression". *Packet Video 2003*.

LEE, K.-H., LEE, S.-H., KO, S.-J. 1998. "Digital image stabilizing algorithms based on bit-plane matching". In *IEEE Transactions on Consumer Electronics*, *Vol. 44*, No. 3, pp. 617-622.

LITVIN, A., KONRAD, J., KARL, W.C. 2003. "Probabilistic Video Stabilization Using Kalman Filtering and Mosaicking". *Proceedings of SPIE Conference on Electronic Imaging*.

MORIMOTO, C.H., CHELLAPPA, R. 1996. “Automatic digital image stabilization”. *Proceedings of IEEE International Conference on Pattern Recognition.*

OLIVIERI, S., HAAN, G., ALBANI, L. 1999. “Noise-robust recursive motion estimation for H.263-based videoconferencing systems”. In *Proceedings of International Workshop on Multimedia Signal Processing, September 1999, Copenhagen*, pp. 345-350.

SHI, J., TOMASI, C. 1999. “Good Features to Track”. In *IEEE Conference on Computer Vision and Pattern Recognition 1999*, pp. 593-600.

SZELISKI, R. 1996. “Video Mosaics for Virtual Environments”. In *IEEE Computers Graphics and Applications*, *Vol. 16*, No. 2, pp. 22-30.

About the authors

Dmitriy Vatolin is a expert in image, video and data compression with more than 12 years experience. Ph.D. in graphics compression, IEEE Member. Books: "Image compression algorithms" (D.Vatolin), "Data compression methods" (D.Vatolin, A.Ratushniak, M.Smirnov, V.Yukin). Senior researcher at the Laboratory of Computer Graphics and Multimedia at Moscow State University, Department of Computational Mathematics and Cybernetics. Head of VideoGroup. Founder of “The Compression Project” http://compression.ru/ and “Compression Links” http://www.compression-links.info/

E-mail: dmitriy@graphics.cs.msu.ru

Stanislav Soldatov is senior researcher at the Laboratory of Computer Graphics and Multimedia at Moscow State University, Department of Computational Mathematics and Cybernetics. E-mail: stass@graphics.cs.msu.ru

Konstantin Strelnikov – Ph.D. student at the Laboratory of Computer Graphics and Multimedia at Moscow State University, Department of Computational Mathematics and Cybernetics. E-mail: kstrelnikov@graphics.cs.msu.ru